



مدارس
للتنولوجيا
التطبيقية

IT Department
Web Development and Programming

تطوير المواقع والبرمجيات

Unit 18 Nodejs

الصف الثالث

Web Development and Programming
Question Bank

1st.

2025 - 2026

تطوير المواقع والبرمجيات

Unit 18 Nodejs

الصف الثالث

Unit 18

Node.js

Unit	18
Name	Node.js
<u>Goals / Outcomes</u>	➤ <u>Remembering</u> 1. Recall core Node.js features and use cases. 2. Remember basic TypeScript syntax and benefits. 3. Identify Jasmine syntax and testing structure. ➤ <u>Understanding</u> 1. Explain Node.js's event-driven, non-blocking I/O. 2. Understand TypeScript's role and benefits in Node.js. 3. Describe Jasmine's testing principles for Node.js. ➤ <u>Applying</u> 1. TypeScript in Node.js projects. 2. Write and run tests with Jasmine in Node.js. 3. Create a Node.js server handling routes and requests. ➤ <u>Analyzing</u> 1. Compare Node.js with other server-side technologies. 2. Assess TypeScript's advantages in development. 3. Resolve Jasmine test failures effectively. ➤ <u>Evaluating</u> 1. Evaluate Node.js server design and performance. 2. Judge the effectiveness and coverage of unit tests. 3. Select the best tools for Node.js development and testing. ➤ <u>Creating</u> 1. Develop a server-side app using Node.js. 2. Create a TypeScript project in Node.js. 3. Build a unit test suite using Jasmine for Node.js.

	4. Develop an interactive website with Node.js.	
Knowledge	Code	Description
	TPK19	Database and data manipulation and Nodejs framework
	TPK20	writing asynchronous code using various techniques
	TPK21	Basic routing and calling a controller method from a route
	TPK22	Analyze and solve common web applications tasks by writing PHP programs
Skill	Code	Description
	TPC5.1	Use database software tools to run queries and produce reports
	TPC5.2	Write PHP scripts to handle HTML forms
	TPC5.5	Creating an authentication system for your application
	TPC5.8	Creating and using composer packages
	TPC5.9	Create restful services, use an Effect Dependency Array and how to hand errors in data requests
Unit Preface		



المعرفة	الكود
Database and data manipulation and Nodejs framework	TPK19

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement.

No	Question	Answer
1	The client is the server or place that contains the backend code.	False
2	Node.js itself is a programming language.	False
3	The V8 engine was developed by Google.	True
4	Node.js is considered a framework like Express.js.	False
5	Netflix and PayPal are examples of companies that use Node.js.	True
6	Node.js is poorly suited for developing real-time applications like chat apps.	False
7	A blocking operation allows tasks to run independently without waiting for each other.	False
8	When you install Node.js, NPM (Node Package Manager) is installed automatically.	True
9	The http module is an example of a third-party module.	False
10	A local module is a module that you create yourself for your own application.	True

Dear learner: Choose the correct answer.

No	Question	Answer
11	Which of the following is NOT one of the three main parts of a website's architecture? a) Client b) Server c) Database d) Framework	D
12	What is Node.js? a) A JavaScript framework b) A new programming language c) A runtime environment for executing JavaScript code d) A web browser	C
13	Which technology, developed by Google, does Node.js leverage to execute its code? a) The Event Loop b) The V8 Engine	B

	c) NPM d) Express.js	
14	Which Node.js architectural feature is responsible for handling asynchronous operations effectively? a) The V8 Engine b) The Module System c) The Event Loop d) The Package Manager (NPM)	C
15	Which tool is primarily used for adding real-time, bidirectional communication to a Node.js application? a) Express.js b) Nest.js c) Koa.js d) Socket.io	D
16	Which of these is a lightweight Node.js framework known for its minimal and flexible design? a) Nest.js b) Socket.io c) Koa.js d) Express.js	D
17	What is a key advantage of using Node.js? a) It uses a single programming language (JavaScript) on both client and server. b) It is the best choice for all CPU-intensive tasks. c) It has no disadvantages. d) It does not require a package manager.	A
18	How would you describe a "Non-Blocking" operation? a) Tasks must complete in order, one after the other. b) Tasks can start and run independently without waiting. c) It prevents other operations from starting. d) It is another term for a synchronous operation.	B
19	How do you verify the version of Node.js installed on your system? a) node -v b) npm -v c) install node d) run node	A
20	The fs (file system) module in Node.js is an example of what? a) A Third-Party Module b) A Local Module c) A Core Module d) A Framework	C

Give the Scientific Name Questions:

No	Question	Answer
21	The part of a website that represents the browser on the computer is called the .	client
22	The server or place that contains the backend code is called the: .	Server
23	The open-source, cross-platform runtime environment for executing JavaScript code is: .	Node.js
24	The software component that transforms web pages into a visual representation is a: .	browser engine
25	The open-source JavaScript and Web Assembly engine developed by Google is called: .	V8
26	A Node.js framework known for its simple design that lets developers create customized applications: is .	Express.js
27	A library that facilitates real-time communication by allowing clients and servers to exchange messages instantly is: .	Socket.io
28	A type of operation where a task must finish before the next one can start is called: .	Blocking
29	A type of operation where tasks can start and continue independently of each other is called: .	Non-blocking
30	The tool that comes with Node.js for managing packages is the: .	npm
31	A type of Modules in node.js is a Self-contained block of code that can be reused throughout an application is called a.	local module
32	Modules that can be included in a program using NPM online, like Express or Mongoose, are called: .	third party modules

Complete the Sentence Questions:

No	Question	Answer
33	A website usually consists of three main parts: the client, the server, and the .	Database
34	The server sends a _____ back to your computer, which can be website content or data for apps.	response
35	Node.js is not a programming language; it is a JavaScript _____.	runtime environment
36	Node.js leverages Google's Chrome _____ for fast and efficient code execution.	V8 Engine
37	The _____ is an architecture in Node.js for handling asynchronous requests effectively.	Event Loop
38	The fs core module is used for working with the _____.	file system
39	Third-party modules are included in a project using the _____ command.	npm install

40 Match the Node.js component with its description.

No	Component	No	Description	Answer
1	Client	A	The place that contains the backend code.	1-C
2	Server	B	Contains all the data that must be saved.	2-A
3	Database	C	Represents the browser on the computer.	3-B
4	V8 Engine	D	A core component that compiles JavaScript code.	4-D
5	Express.js	E	A library for real-time client-server communication.	5-F
6	Socket.io	F	A lightweight framework for building web apps.	6-E

41 Match the term with its correct type.

No	Term	No	Type	Answer
1	Node.js	A	Framework	1-C
2	JavaScript	B	Engine	2-D
3	Express.js	C	Runtime Environment	3-A
4	V8	D	Programming Language	4-B
5	http	E	Third Party Module	5-F
6	mysql	F	Core Module	6-E

Dear Learner: Answer the following questions.

42	Question	What are the three main parts of a website's architecture?
	Answer	The client, the server, and the database.

43	Question	Define: Node.js?
	Answer	Node.js is an open-source, cross-platform runtime environment for executing JavaScript code outside a web browser.

44	Question	What is the key difference between blocking and non-blocking operations?
	Answer	Answer: Blocking operations must be completed before the next task can start, while non-blocking operations can run independently without waiting.

45	Question	What command do you use to check if Node.js is installed correctly on your system?
	Answer	node -v

المعرفة	الكود
writing asynchronous code using various techniques	TPK20

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement.

No	Question	Answer
46	TypeScript code is executed directly by a web browser's JavaScript engine.	False
47	All valid JavaScript code is automatically valid TypeScript code.	True
48	The tsconfig.json file is used to configure how the TypeScript compiler behaves.	True
49	In TypeScript, a tuple is an array with a fixed length where each element can be a different type.	True
50	A class is an instance of an object.	False
51	The private access modifier allows a class property to be accessed from anywhere in your code.	False
52	TypeScript is considered a superset of JavaScript.	True
53	Inheritance allows a subclass to inherit properties and methods from a superclass.	True
54	Polymorphism means that a class can have multiple methods with the same name but different parameters.	False
55	You can initialize a TypeScript project using the command <code>npx tsc --init</code> .	True

Dear learner: Choose the correct answer.

No	Question	Answer
56	Who developed and maintained TypeScript? A) Google B) Mozilla C) Microsoft D) Apple	c
57	Which of these is a primary benefit of using TypeScript over JavaScript? A) Code runs faster in the browser B) It doesn't need to be compiled C) Early error detection through static typing D) It has a completely different syntax	c
58	Which command is used to install TypeScript globally using npm? A) <code>npm install typescript</code>	d

	B) npm add -g typescript C) node install typescript D) npm install -g typescript	
59	What is the correct file extension for a TypeScript source file? A) .ts B) .txt C) .js D) .typescript	a
60	Which TypeScript type would you use to represent a value that could be either true or false? A) string B) number C) boolean D) tuple	c
61	Which OOP concept is achieved in TypeScript by using the private and protected keywords? A) Inheritance B) Encapsulation C) Polymorphism D) Abstraction	b
62	If a class member is marked as protected, where is it accessible? A) Anywhere B) Only within its own class C) Within its class and its subclasses D) Only outside its class	c
63	What is the output of the TypeScript compiler? A) JavaScript files (.js) B) TypeScript bytecode C) Machine code D) A single executable file	a
64	What is the purpose of Generics in TypeScript? A) To generate random numbers B) To create reusable, type-safe components C) To define the generic structure of an object D) To handle errors	b
65	The ability for a subclass to provide a specific implementation of a method that is already defined in its superclass is known as: A) Encapsulation B) Overriding (a form of Polymorphism) C) Abstraction D) Inheritance	b

Give the Scientific Name Questions:

No	Question	Answer
66	A language that builds on JavaScript by adding static typing is called:	TypeScript
67	The Microsoft-developed open-source language that is a superset of JavaScript is:	TypeScript
68	The principle of hiding internal implementation details and controlling access to properties is known as:	Encapsulation
69	A user-defined type that restricts values to a set of named constants is called an:	enum
70	A fixed-length array where each position can have a different type is called a:	tuple
71	The JSON file that defines settings for the TypeScript compiler is named:	tsconfig.json
72	The process where a subclass provides a different implementation of a method from its superclass is called _____.	Polymorphism
73	An _____ is a blueprint for creating objects, defining their properties and methods.	class
74	The access modifier that makes a property accessible only within its own class is :	private
75	The process of hiding implementation details and showing only essential features is known as:	Abstraction

Complete the Sentence Questions:

No	Question	Answer
76	TypeScript is a _____ of JavaScript developed by Microsoft.	superset
77	TypeScript uses _____ typing, which helps catch errors early. Answer:	static
78	TypeScript files use the _____ file extension.	.ts
79	The TypeScript compiler converts .ts files into _____ files that browsers can run.	.js
80	The number type represents both integers and _____.	decimals
81	An array defined as let list: string[] can only contain values of type _____.	string
82	In a tuple [string, number], the first element must be a _____.	string
83	The _____ keyword is used to define a class in TypeScript.	class
84	A property marked as _____ is only accessible within the class it is defined in.	private
85	_____ allows a subclass to inherit properties and methods from a superclass.	Inheritance

86 Match the TypeScript concept to its description.

No	Concept	No	Description	Answer
1	Interface	A	A fixed-length array with pre-defined types for each index.	1-B
2	Tuple	B	Defines the structure of an object for type safety.	2-A
3	Enum	C	An instance of a class.	3-D
4	Object	D	A user-defined set of named constants.	4-C
5	Generics	E	The blueprint for creating objects.	5-F
6	Class	F	Create reusable components that work with multiple types.	6-E

Dear Learner: Answer the following questions.

87	Question	What is the primary technical difference between TypeScript and JavaScript?
	Answer	TypeScript is statically typed, while JavaScript is dynamically typed.

88	Question	What is the purpose of the tsconfig.json file?
	Answer	To configure the behavior of the TypeScript compiler (defines compiler settings).

89	Question	What command do you use to compile a TypeScript file named app.ts?
	Answer	tsc app.ts

90	Question	What are the four main principles of Object-Oriented Programming supported by TypeScript?
	Answer	Encapsulation, Inheritance, Polymorphism, Abstraction.

المعرفة	الكود
Basic routing and calling a controller method from a route	TPK21

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement.

No	Question	Answer
91	Jest was originally developed by Google.	False
92	Unit tests are designed to check how different parts of an application work together.	False
93	Jest has a built-in test runner that can run tests in parallel.	True
94	Jasmine's popularity increased from 2016 to 2020.	False
95	Snapshot tests are primarily used to test the business logic of a function.	False
96	Integration tests build upon the foundation of unit tests.	True
97	To install Jest, you use the command <code>npm install --save-dev jest</code> .	True
98	Jest provides support for both TDD and BDD.	True
99	The primary goal of a unit test is to verify that a function always returns the expected output for a given input.	True
100	Code coverage is a feature that helps identify which parts of the code need more testing.	True

Dear learner: Choose the correct answer.

No	Question	Answer
101	What is the main purpose of unit testing? a) To test how different services integrate b) To check the smallest parts of code, like a single function c) To verify the appearance of the user interface d) To test the entire application from start to finish	b
102	Which company developed the Jest testing framework? a) Google b) Microsoft c) Facebook (Meta) d) Netflix	c
103	Which of the following is a key advantage of Jest according to the text? a) It requires many complex configuration files. b) It must be used with an external test runner like Karma. c) It is easy to set up and has a simple configuration. d) It is only designed for testing Angular applications.	c

104	What is the correct naming convention for a Jest test file? a) test.add.js b) add.js.test c) *.test.js or *.spec.js d) jest-test.add.js	c
105	Snapshot testing is most useful for: a) Verifying mathematical calculations in a function b) Checking the logical flow of an API endpoint c) Ensuring the rendered UI does not change unexpectedly d) Testing database connection speeds	c
106	What does the expect(add(1, 2)).toBe(3); assertion do? a) Mocks the add function to always return 3 b) Checks if the function add exists c) Verifies that calling add(1, 2) returns the value 3 d) Measures the execution speed of the add function	c
107	Which development methodologies does Jest specifically support? a) Waterfall and Agile b) TDD and BDD c) Scrum and Kanban d) Spiral and V-Model	b
108	The practice of replacing real objects with test versions to isolate the code under test is called: a) Snapshotting b) Running c) Mocking d) Compiling	c
109	According to the text, what was a major reason for Jasmine's decline in popularity? a) It became too expensive to use. b) It doesn't support newer features like snapshots and often requires extra tools. c) It only works with the React framework. d) It was acquired by Facebook and merged into Jest.	b
110	What do integration tests typically check? a) The performance of a single algorithm b) The syntax of the code c) How different units or components work together d) The color scheme of the application	c

Complete the Sentence Questions:

No	Question	Answer
111	A popular JavaScript testing framework developed by Facebook is called _____.	Jest
112	The type of testing that focuses on the smallest parts of code, like a single function, is called _____ testing.	unit
113	_____ is a development approach where tests are written based on the desired behavior of the code.	BDD
114	A testing feature that saves a copy of the UI output to compare against future renders is called _____ testing.	snapshot
115	The development approach where you write tests before writing the actual code is known as _____.	TDD
116	Jest's ability to replace real objects with simplified, controllable versions for testing is known as _____.	mocking
117	Tests that check how different parts of an application work together are called _____ tests.	integration
118	The older JavaScript testing framework that often requires extra tools like Karma is called _____.	Jasmine
119	The naming convention for Jest test files is either *.test.js or *._____.js.	spec
120	The command to install Jest as a development dependency is npm install --save-dev _____.	jest

المعرفة	الكود
Analyze and solve common web applications tasks by writing PHP programs	TPK22

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement.

No	Question	Answer
121	Express.js is a front-end JavaScript framework.	False
122	npm is used to execute JavaScript in a web browser.	False
123	Middleware functions handle requests and responses between the client and server.	True
124	The package.json file is automatically created when you run the command npm install express.	False
125	The primary purpose of Nodemon is to compile TypeScript code.	False
126	In the provided API, a GET request to /api/employees/1 would retrieve the employee with an ID of 1.	True

127	A PUT request is used specifically for deleting records in a REST API.	False
128	The app.use(express.json()) middleware is necessary to parse JSON data from the request body.	True
129	The req.params object is used to access data sent in the URL path (like an ID).	True
130	The req.body object is used to access data sent in the request's JSON body.	True

Dear learner: Choose the correct answer.

No	Question	Answer
131	What is the primary purpose of Express.js? a) To style web pages b) To build server-side web applications and APIs c) To manage client-side state d) To create database schemas	b
132	Which command is used to initialize a new Node.js project and create a package.json file? a) npm start b) node init c) npm install express d) npm init	d
133	Which of the following is NOT a feature of Express.js mentioned in the text? a) Middleware b) Template Engines c) Virtual DOM d) Error Handling	c
134	In the context of the provided CRUD API, which HTTP method is correctly paired with its operation? a) POST - Delete an employee b) GET - Update an employee c) PUT - Create a new employee d) DELETE - Delete an employee	d
135	Which middleware function must be used to parse incoming JSON request bodies? a) app.use(express.static()) b) app.use(express.urlencoded()) c) app.use(express.json()) d) app.use(express.router())	c
136	How is a specific employee's ID typically accessed from the URL in an Express route like /employees/:id?	c

	a) req.query.id b) req.body.id c) req.params.id d) req.url.id	
137	What does API stand for? a) Application Program Interaction b) Applied Programming Interface c) Application Programming Interface d) Automated Program Interaction	c
138	Which tool is suggested for testing the API endpoints? a) Microsoft Word b) A web browser only c) Postman d) MongoDB Compass	c
139	What does CRUD stand for? a) Create, Read, Update, Delete b) Call, Render, Update, Destroy c) Create, Render, Use, Deploy d) Calculate, Return, Use, Define	a
140	According to the analogy in the text, an API is likened to: a) A restaurant menu b) A chef cooking food c) A waiter taking an order d) A restaurant guest	c

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement.

No	Question	Answer
141	NoSQL databases are primarily designed for managing financial transactions.	False
142	A MongoDB collection is conceptually equivalent to an SQL table.	True
143	Template engines like EJS mix application logic with the presentation layer, making code less maintainable .	False
144	In EJS, the <code><% %></code> syntax is used to output the value of a JavaScript expression into the HTML.	False
145	Middleware in Express.js can only run after a route handler has processed a request.	False
146	CORS is a security feature enforced by web servers, not web browsers.	False
147	CORS is designed to make cross-origin requests easier and less secure.	False

148	In the context of databases, a field in a MongoDB document is analogous to a column in an SQL table.	True
149	The main purpose of a template engine is to style HTML pages with CSS.	False
150	The res.render() method in an Express.js application is used to process an EJS template with data and send the resulting HTML.	True

Dear learner: Choose the correct answer.

No	Question	Answer
151	Which of these is a primary use case for an SQL database? A) Social Media Platforms B) Real-time Analytics C) Financial Transactions D) Content Management Systems with User-Generated Content	c
152	In MongoDB, what is the equivalent data structure to an SQL row? A) Database B) Collection C) Field D) Document	d
153	What is the primary benefit of using a template engine like EJS? A) It compiles JavaScript into machine code. B) It manages database connections efficiently. C) It separates the presentation layer (HTML) from the application logic (JavaScript). D) It enforces CORS policies on the server.	c
154	Which EJS tag is used to output the value of a variable into the final HTML? A) <%= %> B) <% %> C) <% - %> D) <%# %>	a
155	What is the role of middleware in an Express.js application? A) To define the database schema B) To act as a layer that intercepts and manipulates requests and responses C) To compile EJS templates into HTML D) To style the front-end components	b
156	The main purpose of CORS is to: A) Prevent all access to a web server. B) Securely allow web pages to make requests to a different domain.	b

	C) Encrypt data stored in a database. D) Improve the performance of template rendering.	
157	Which of the following is a NoSQL document database? A) MySQL B) PostgreSQL C) Microsoft SQL Server D) MongoDB	d
158	In the provided EJS example, what does the variable users represent inside the template? A) A JavaScript function to query the database. B) The data passed from the Node.js application to the template. C) A built-in EJS keyword for creating loops. D) The name of the MongoDB collection.	b
159	What does the app.set('view engine', 'ejs') line of code do? A) It installs the EJS module from npm. B) It tells Express.js to use EJS as its default template engine. C) It creates a new EJS template file. D) It sends an EJS file directly to the client.	b
160	Which is a NoSQL use case? A) Financial Transactions B) Enterprise Resource Planning (ERP) C) Social Media Platforms D) E-commerce Product Catalogs	c

Give the Scientific Name Questions:

No	Question	Answer
161	A popular web framework built on top of Node.js that simplifies building web applications and APIs is called:	Express.js
162	Functions that handle requests and responses between the client and server are known as:	Middleware
163	The acronym for an interface that connects different applications is:	API
164	Tools like EJS and Pug that help separate the presentation layer from the application logic are called:	Template Engines
165	The security mechanism that prevents web pages from making requests to a different domain is called:	CORS
166	The set of operations (Create, Read, Update, Delete) used to manage data is known as:	CRUD
167	In the context of databases, a single record in a MongoDB collection is called a :	document

168	The file that stores all project info and dependencies for a Node.js application is:	package.json
169	The method used to send a JSON response in an Express.js route handler is	.json(). Or res.json()

170 Match the HTTP method to the CRUD operation it typically performs.

No	HTTP Method	No	CRUD Operation	Answer
1	GET	A	Delete	1-B
2	POST	B	Read	2-C
3	PUT	C	Create	3-E
4	PATCH	D	Update (Partial)	4-D
5	DELETE	E	Update (Full)	5-A
6	All of them	F	CRUD	6-F

Dear Learner: Answer the following questions.

171	Question	What is the primary purpose of Express.js?
	Answer	To simplify the process of building web applications and APIs in Node.js.

172	Question	What is the role of middleware?
	Answer	To handle and process requests and responses between the client and the server.

173	Question	What is the key structural difference between SQL and NoSQL databases?
	Answer	SQL uses tables with fixed rows and columns, while NoSQL uses flexible documents or other models.

174	Question	What is the main benefit of using a template engine like EJS?
	Answer	It separates the presentation layer (HTML) from the application logic (JavaScript).

175	Question	What command is used to start an Express server with automatic restarts on file changes the file called server.js?
	Answer	nodemon server.js / npx nodemon server.js